

Development of the embryo, larva and early juvenile of Nile tilapia *Oreochromis niloticus* (Pisces: Cichlidae). Developmental staging system

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We described the developmental stages for the embryonic, larval and early juvenile periods of Nile tilapia *Oreochromis niloticus* to elucidate sequential events of craniofacial development. Craniofacial development of cichlids, especially differentiation and morphogenesis of the pharyngeal skeleton, progresses until about 30 days postfertilization (dpf). Because there is no comprehensive report describing the sequential processes of craniofacial development up to 30 dpf, we newly defined 32 stages using a numbered staging system. For embryonic development, we defined 18 stages (stages 1–18), which were grouped into seven periods named the zygote, cleavage, blastula, gastrula, segmentation, pharyngula and hatching periods. For larval development, we defined seven stages (stages 19–25), which were grouped into two periods, early larval and late larval. For juvenile development until 30 dpf, we defined seven stages (stages 26–32) in the early juvenile period. This developmental staging system for Nile tilapia *O. niloticus* will benefit researchers investigating skeletogenesis throughout tilapia ontogeny and will also facilitate comparative evolutionary developmental biology studies of haplochromine cichlids, which comprise the species flocks of Lakes Malawi and Victoria.

Key words: cichlid, craniofacial morphology, developmental stage, Nile tilapia *Oreochromis niloticus*.

Introduction

One of the most fundamental problems in developmental biology is the standardization of animal development. In this respect, a staging series is a useful standardization tool that provides accuracy in developmental studies. Work describing developmental stages has been published, not only for tetrapods (e.g. chicken, Hamburger & Hamilton 1951; frog, Nieuwkoop & Faber 1994; mouse, Theiler 1989), but also for fish (e.g. lamprey, Tahara 1988; dogfish, Ballard *et al.* 1993; zebrafish, Kimmel *et al.* 1995; rainbow trout, Ballard 1973; stickleback, Swarup 1958; medaka, Iwamatsu 2004). Developmental staging allows us to compare the developmental processes between the model species and related species to reveal underlying mechanisms of evolutionary changes among them.

Cichlids (Cichlidae; Perciformes) are a family of the most successful fish among teleosts. It has been estimated that the number of their species is more than 1350 in the world (Nelson 2006). It is said that several hundred more remain undiscovered and probably exist in Africa (Turner *et al.* 2001). In East Africa, Lakes Victoria, Malawi and Tanganyika harbor several hundred endemic species of cichlids (Seehausen 1996; Turner *et al.* 2001). These cichlids are famously spectacular examples of explosive adaptive radiation and exhibit remarkable morphological, behavioral and ecological diversity (Frayser & Iles 1972). In particular, the cichlids of Lakes Malawi and Victoria, which are called haplochromine, have evolved within a short period and thus have attracted much attention among evolutionary biologists (Kocher 2004).

The haplochromine cichlids exhibit a vast diversity of feeding habits. The morphologies of the feeding apparatus, including the jaw and teeth, have surprising diversity due to adaptation to their feeding behavior (Frayser & Iles 1972; Liem 1991). Therefore, a general question is raised: How have the morphologies of the feeding apparatus evolved during adaptation? To answer this question, we need to first describe the developmental processes for the morphologies of the

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cichlid feeding apparatus. Craniofacial development, including the feeding apparatus, is a highly regulated system involving control of specification and morphogenesis for the head skeleton. As such, standardization of embryonic timing is required for adequate research on evolutionary developmental biology. However, there is not yet a comprehensive staging system that encompasses a wide range of cichlid craniofacial development, although there have been many studies describing the developmental stages of cichlid fish.

Nile tilapia *Oreochromis niloticus* is one of the most important and useful species of cichlids, not only for fresh-water aquaculture (FAO 2004), but also for research in physiology (Wright & Land 1998), endocrinology (Melamed *et al.* 1998; Seale *et al.* 2002), genomic biology and molecular genetics (Majumdar & McAndrew 1986; Kocher *et al.* 1998; Oliveira & Wright 1998; McConnell *et al.* 2000; Lee *et al.* 2003; Katagiri *et al.* 2005; Lee *et al.* 2005; Santini & Bernardi 2005). Molecular technology using transgenic fish of *O. niloticus* has also been established (Maclean *et al.* 2002). *O. niloticus* has a high fecundity (hundreds of eggs in a clutch), year-round breeding capability (2–3-week spawning cycle), and a short generation period (half a year). These characteristics make *O. niloticus* an excellent model species for studying evolutionary developmental biology in cichlids.

Several studies have assessed the developmental stages of *O. niloticus* (Nussbaum & Chervinsky 1968; Galman 1980; Lingling & Qianru 1981; Galman & Avtalion 1989; Rana 1990; Morrison *et al.* 2001). Each author chose different features for staging under different rearing temperatures and proposed a different staging system. Of these works, the study by Morrison *et al.* (2001), which used the same rearing temperature ($28 \pm 1^\circ\text{C}$) as we chose in the present study, detailed the development of the embryo and early larva of *O. niloticus*. The Morrison *et al.* study has two elements of interest. First, the authors proposed a named, but not numbered, staging system that followed the zebrafish staging system (Kimmel *et al.* 1995). Second, they primarily provided information regarding internal organs and some glands using light micrographs of sectioned material from the embryos and larva until swimbladder inflation and exogenous feeding. They evaluated development with respect to internal organs (only until 9 days postfertilization (dpf)) but provided little information regarding the skeleton.

We considered the construction of a new staging system for cichlids to be necessary for four reasons: (1) no detailed description of craniofacial development in *O. niloticus* has been presented until now; (2) easy recognition of the sequential events during late development is required because the description of

late developmental processes using developmental time, body length, or named stage is sometimes ambiguous; (3) it is important to compare the developmental process of *O. niloticus* to other cichlids or other model species; and (4) it is necessary to be able to easily recognize the external criteria of a particular stage through a dissecting microscope so that molecular biological experiments, such as RNA extraction or *in situ* hybridization, can be conducted. In the first of two papers, we described the developmental stages of *O. niloticus* to elucidate sequential events of craniofacial development. In the second paper, we will describe the head skeletogenesis of *O. niloticus*.

Materials and methods

Nile tilapia, *Oreochromis niloticus*

Embryos and juveniles of *O. niloticus* that were obtained as genetic females (XX) and males (XY) (Kobayashi *et al.* 2000) were a gift from Dr Tohru Kobayashi and Dr Yoshitaka Nagahama of the National Institute of Basic Biology (Okazaki, Japan). The embryos and juveniles were transferred to the Tokyo Institute of Technology (Yokohama, Japan) and raised to adult fish for almost 1 year. Each of the adult fish was maintained under constant conditions ($28 \pm 1^\circ\text{C}$, 10 h dark/14 h light cycle) in 100 L tanks through which fresh water was circulated, and fed three times daily with granulated commercial foods.

To obtain specimens used in the present study, artificial fertilizations using 1- to 2-year-old adult fish were performed. Females ready for spawning were identified by protruding genital papilla and when they began digging into the small pebbles in the tanks with their mouths to generate circular nests. Each female and male was anesthetized in 0.02% ethyl *p*-aminobenzoate (Wako, Osaka, Japan) and gently stripped to obtain eggs and sperm, respectively. The sperm suspension and eggs were gently mixed for 5–10 min in a Petri dish and then washed several times with the system water. Thus, we obtained several hundreds of fertilized eggs per an artificial fertilization.

The embryos were cultured in round-bottom bottles in to which the system water was gently supplied through the tank system. We usually kept < 200 embryos in a 50 mL bottle. Around 8 dpf, and before the larvae began to float via swimbladder inflation, the larvae were moved to 1 L containers through which the tank system water was supplied. We usually kept < 300 larvae in a 1 L container. After 12 dpf, when the juveniles could freely swim and were able to forage for baby brine shrimp, the juveniles were

moved to 40 L tanks in the tank system. We usually kept < 300 juveniles in a 40 L tank when the juveniles were moved. As the fish grew up, we thinned out the population in a tank. For 1 month, the juveniles were fed with baby brine shrimp and then gradually with powdered commercial foods, taking care to maintain optimal growth and density of the fish in the tanks. We eliminated unfertilized eggs and embryos that exhibited deformity during their development. In a clutch, we observed approximately 5% of eggs as unfertilized and approximately 5–10% of embryos and larvae as deformity, depending on conditions of the mother fish or the system water. When we observed more frequently (> 30%), we did not use the clutch.

Observation and photographs

Specimens were killed or anesthetized using 0.02% ethyl-3-aminobenzoate methanesulfonate salt (Sigma, St. Louis, MO, USA). The specimens were mounted on a depression slide without an overlying coverslip in 3% methyl cellulose (Sigma) to allow positioning as desired. The lower magnification views showing the entire specimens were made using a Leica MZ16 FA dissecting stereomicroscope (Leica Microsystems GmbH, Wetzlar, Germany) with an Olympus DP70 CCD camera (Olympus, Tokyo, Japan). The higher magnification views, showing parts of the living specimens, were taken with Nomarski differential interference contrast (DIC) optics of a Carl-Zeiss Axioplan with a Carl-Zeiss AxioCam CCD camera (Carl-Zeiss AG, Oberkochen, Germany). We used 10× and 20× objectives, which reveal details in living specimens. The images in Figures 4C,I,M and 5E were generated from several DIC images and processed using Photoshop 6 (Adobe Systems, San Jose, CA, USA).

To better visualize the embryos at 4 and 5 dpf (stages 14–16), the chorions were removed easily, probably due to endogenous hatching enzyme produced in the hatching glands, using sharp forceps. However, the chorions could not be easily removed from embryos at 1–3 dpf due to large and soft yolks. We then observed the specimens as follows: (1) the living specimens in stages 1–11 were untreated and with chorions (Figs 1A–E and 4C,D); (2) the specimens in stages 10–13 were fixed (see below) (Figs 3A–J and 4A,E,G,K), dechorionated and stained (see below) (Figs 3K and 4B,F,H,J); (3) the living specimens in stages 12 and 13 were treated with pronase or hatching medium for dechoriation (Morrison *et al.* 2003) (Fig. 4I,J,M).

To better observe the embryos that had large yolks during cichlid embryogenesis (stages 10–13), the specimens with chorions were observed after fixation

with 4% paraformaldehyde (PFA; Wako) in phosphate-buffered saline (PBS) (PFA/PBS), followed by treatment with Bouin's fixative (Hopwood 1990; Fournie *et al.* 2000) (Figs 3A–J and 4A,E,G,K). Staining is required for further observation of cichlid's embryogenesis. Hematoxylins are the most widely used stains for teleost embryology because they stain the yolk only slightly (Galat 1972). Modified from Galat (1972), some of the fixed embryos were dechorionated and stained with Mayer's hematoxylin solution (Wako) for 5–10 min and washed with deionized water several times (Figs 3K and 4B,F,H,L). Mayer's hematoxylin stains the nuclei a red color (the color would be converted to blue-black if the specimens were washed in weak alkali) (Stevens 1990).

Length records were obtained from living specimens. About 1200 embryos, larvae and juveniles were measured under a dissecting microscope and the records were given as the mean $\pm$ SD. Age was recorded in hours postfertilization (hpf) and dpf, counting the time of fertilization as 0 h and the day of fertilization as the first day.

Results

Overview of the embryonic, larval and early juvenile developmental stages of Nile tilapia *O. niloticus*

We defined the stages of embryonic, larval and early juvenile development of Nile tilapia *O. niloticus*. The 32 stages that we defined are shown in Figure 1. The characteristics of each developmental stage are shown in Table 1. The numbered staging system of *O. niloticus* was modified from two different staging systems that were previously described; stages 1–19 were modified from the description by Galman and Avtalion (1989). Stages 20–32 were defined using the number of elements in central caudal fin rays from the description by Otten (1982). The number of caudal fin ray elements (CFRE) could easily be determined through a dissecting microscope under low magnification, using a device for oblique illumination and on a dark field (Fig. 2A). An increase in the number of CFRE during development corresponded with an increase in the body size (black circles in Fig. 2B).

As Morrison *et al.* (2001) mentioned, we also noticed some variations in the rate of development among embryos in the same clutch, as well as among different clutches. Thus, we studied the embryos, larvae and juveniles from several tens of clutches at various developmental stages to ensure our proposed staging system as a standard. We also observed that such variations became prominent especially during the late developmental stages. Fish were well-developed

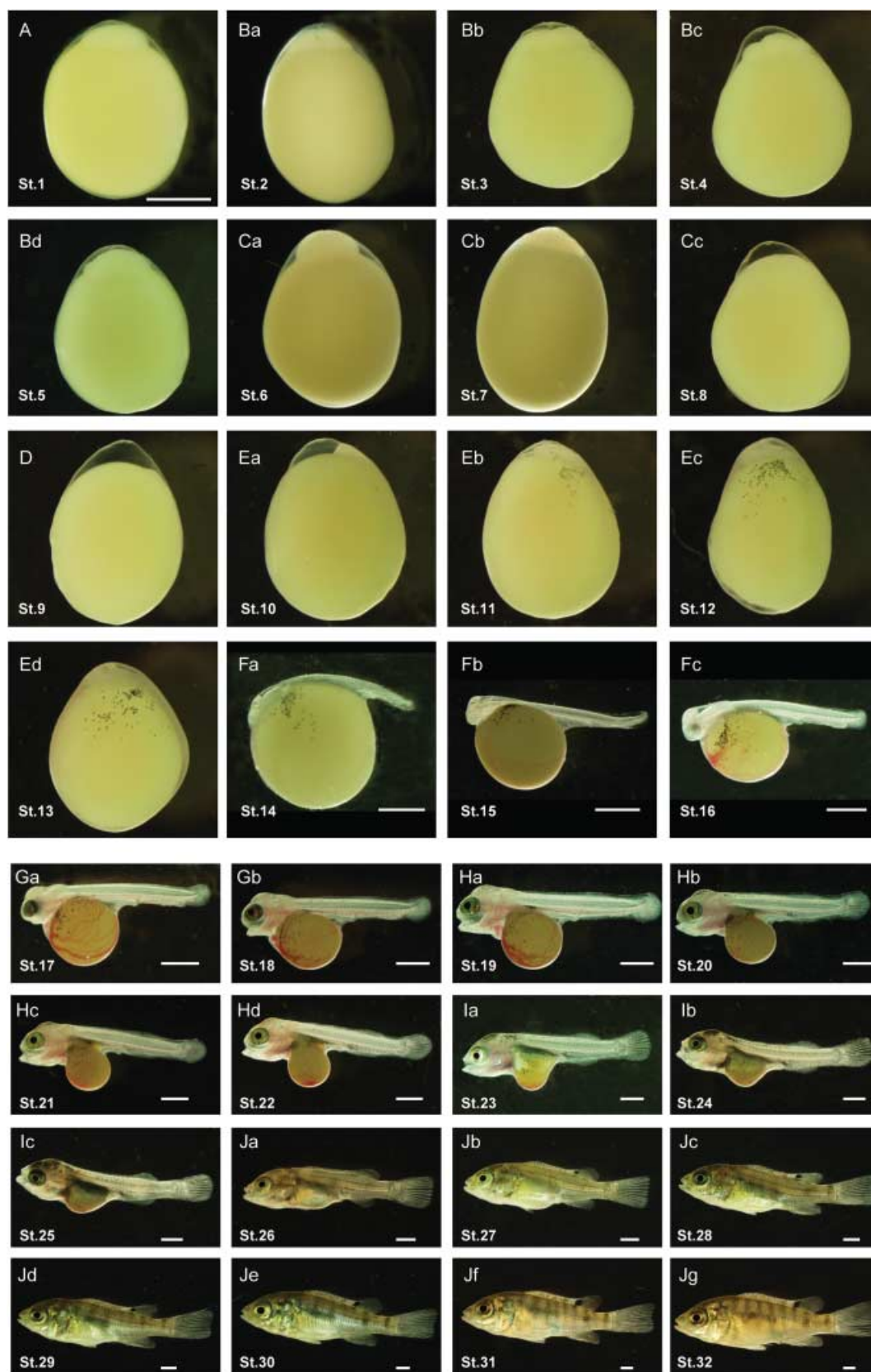


Table 1. Developmental stages of Nile tilapia *Oreochromis niloticus*

Period		Stage	dpf	hpf	Characteristics
Embryo	Zygote	1	1	0–1.5	1-cell
	Cleavage	2	1	1.5–2	2-cell
		3	1	2	4-cell
		4	1	3	8-cell
		5	1	4	16-cell
	Blastula	6	1	4–12	Early blastula
		7	1	12–17	Middle blastula
		8	1	17–22	Late blastula
	Gastrula	9	2	22–26	Gastrula, epiboly = 30–50%
	Segmentation	10	2	26–30	Neurula, epiboly = 50–90%
		11	2–3	30–40	Yolk plug closure and somitogenesis
		12	3	40–44	Optic cups and tail undercut
		13	3	44–48	Brain differentiation
	Pharyngula	14	3–4	48–60	Heart beat
		15	4	60–72	Onset of blood circulation
		16	4–5	72–90	Head enlargement
	Hatching	17	5–6	90–110	Jaw extension
		18	6	110–120	Gill formation
Larva	Early larva	19	6–7	120–132	Opercular movement
		20	7	132–144	CFRE 2; LL = 6.5 ± 0.2 mm
		21	7–8		CFRE 3; LL = 6.9 ± 0.2 mm
		22	8–9		CFRE 4; LL = 7.2 ± 0.3 mm
	Late larva	23	9–10		CFRE 5; TL = 8.1 ± 0.2 mm
		24	9–10		CFRE 6; TL = 8.6 ± 0.2 mm
Juvenile	Early juvenile	25	11–13		CFRE 7; TL = 9.0 ± 0.4 mm
		26	12–13		CFRE 8; TL = 9.9 ± 0.4 mm
		27	14–15		CFRE 9; TL = 10.4 ± 0.4 mm
		28	16–20		CFRE 10; TL = 11.9 ± 0.4 mm
		29	19–22		CFRE 11; TL = 12.9 ± 0.5 mm
		30	21–24		CFRE 12; TL = 14.4 ± 0.5 mm
		31	23–25		CFRE 13; TL = 15.6 ± 0.6 mm
		32	25–30		CFRE 14; TL = 17.3 ± 0.8 mm

dpf, day(s) postfertilization; hpf, hour(s) postfertilization; LL, larval length; TL, total length; CFRE, caudal fin ray elements.

when the adequate density of fish per tank and the proper quantity of food per fish (black circles in Fig. 2B) were provided, while slowly developed fish appeared to be mainly caused by the density of fish and the poor quantity of food (gray squares in Fig. 2B). We obtained the information that represents the optimal development from well-developed fish of each developmental stage (black circles in Fig. 2B).

Following the staging system of Morrison *et al.* (2001) with some modification, we grouped 32 stages into 10 time blocks or periods.

Fig. 1. Overview of the developmental stages of Nile tilapia *O. niloticus*. (A) Stage 1 in zygote period. (B, a–d) Stages 2–5 in cleavage period. (C, a–c) Stages 6–8 in blastula period. (D) Stage 9 in gastrula period. (E, a–d) Stages 10–13 in segmentation period. (F, a–c) Stages 14–16 in pharyngula period. (G, a–b) Stages 17–18 in hatching period. (H, a–d) Stages 19–22 in early larval period. (I, a–c) Stages 23–25 in late larval period and (J, a–g) stages 26–32 in early juvenile period. St, stage. See the criteria for each stage in Table 1. Scale bar, 1 mm.

Embryonic development began with fertilization and ended with hatching out from the chorion (stages 1–18). Embryonic development was subdivided into seven periods: zygote (Fig. 1A), cleavage (Fig. 1B), blastula (Fig. 1C), gastrula (Fig. 1D), segmentation (Fig. 1E), pharyngula (Fig. 1F), and hatching (Fig. 1G). The zygote period (0–1.5 hpf) consisted of one stage, stage 1 (Fig. 1A), and was characterized by cytoplasmic accumulation to form the blastodisc at the animal pole and the appearance of a distinct perivitelline space. The cleavage period (1.5–4 hpf) consisted of four stages; stages 2–5 (Fig. 1B,a–d). The cleavage period was characterized by a series of mitotic divisions that resulted in many blastomeres. The blastula period (4–22 hpf) consisted of three stages; stages 6–8 (Fig. 1C,a–c). The blastula period was characterized by two distinct layers of the blastoderm, an outer enveloping layer and a peripheral yolk syncytial layer (Fleig 1993; Fishelson 1995a). The gastrula period (22–26 hpf) consisted of one stage; stage 9 (Fig. 1D), and was characterized by

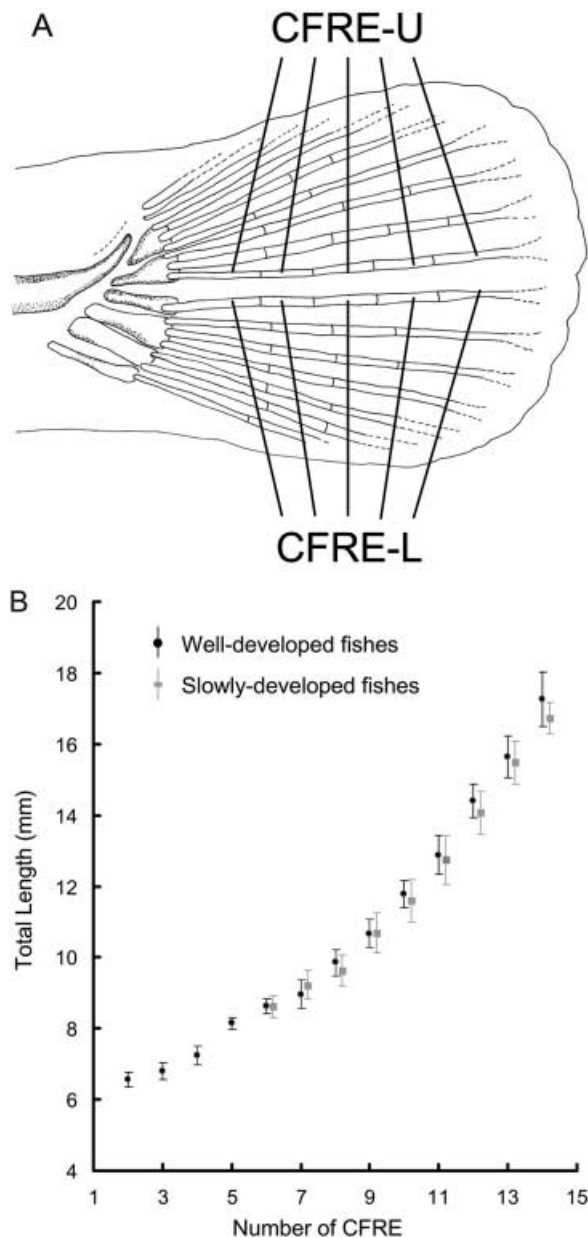


Fig. 2. Number of caudal fin ray elements (CFRE) is an important distinguishing characteristic in later developmental stages. (A) Schematic presentation of the caudal fin of an *O. niloticus* specimen. The central rays could be distinguished and divided into upper and lower parts of the caudal fin. When the number of elements in the upper (CFRE-U) and lower (CFRE-L) central rays of the caudal fin were the same, we determined that as the number of the CFRE. There are five elements in both the upper and lower fin; in this example, the CFRE is five, and thus the specimen belongs to stage 23. (B) Relationship between number of CFRE and total length. An increase in the number of CFRE accompanied an increase in the body size, which represented the development of *O. niloticus*. Each value represents the mean length and the standard deviation from several clutches. The black circles in the graph were obtained from the well-developed fish and gray squares were obtained from the slowly developed fish.

the presence of the germ ring around the margin of the blastoderm and the embryonic shield, which extended from the germ ring toward the animal pole to form the neural tube. The segmentation period (26–48 hpf) consisted of four stages; stages 10–13 (Fig. 1E,a–d). The segmentation period was characterized by somite formation and the partitioned brain. The pharyngula period (48–90 hpf) consisted of three stages; stages 14–16 (Fig. 1F,a–c). The pharyngula period was characterized by the primordia of the pharyngeal arches, which were present, but difficult to distinguish individually at earlier times. The hatching period (90–120 hpf) consisted of two stages; stages 17 and 18 (Fig. 1G,a,b). Most *O. niloticus* eggs hatched out during stage 17 (5 dpf), although some eggs hatched out earlier at stage 15 (4 dpf) or later at stage 18 (6 dpf). The hatching period was characterized by morphogenesis of the pharyngeal skeleton.

Larval development began after the hatching period and we defined the end as the yolk absorption (stages 19–25). Larval development was subdivided into two periods: early larval and late larval. The early larval period consisted of four stages; stages 19–22 (Fig. 1H,a–d). The early larvae gradually began to move its jaws, opercular flaps and pectoral fins. The late larval period consisted of three stages; stages 23–25 (Fig. 1I,a–c). The late larval period was characterized by inflation of the swimbladder and functionalization of the pharyngeal skeleton prior to initiating exogenous feeding.

Juvenile development begins with yolk absorption and lasts until the first maturation of gametes. In the present paper, only the early stages of juvenile development, referred to as the early juvenile period, were defined as stages 26–32 (Fig. 1J,a–g). The early juvenile period was characterized by the onset of exogenous feeding and the appearance of a 'tilapia mark' (or 'tilapia spot'), that is, tilapia-specific juvenile coloration in the middle of the dorsal fin (Barel *et al.* 1977; Holden & Bruton 1992). In the subsequent late juvenile period, the tilapia mark gradually disappeared around 100 dpf, probably accompanying the maturation of gametes (Nakamura & Nagahama 1985, 1989). In general, male fish develop faster than female fish.

Developmental stages in the gastrula and segmentation periods of *O. niloticus*

Because our study was conducted to elucidate the staging of craniofacial development, we first described stages 9–13, the period of development that follows gastrulation and leads immediately to the appearance of the anterior region in the embryos. Like the blastodisc in the blastula period, it was difficult to distinguish the blastoderm in the gastrula period and the embryo

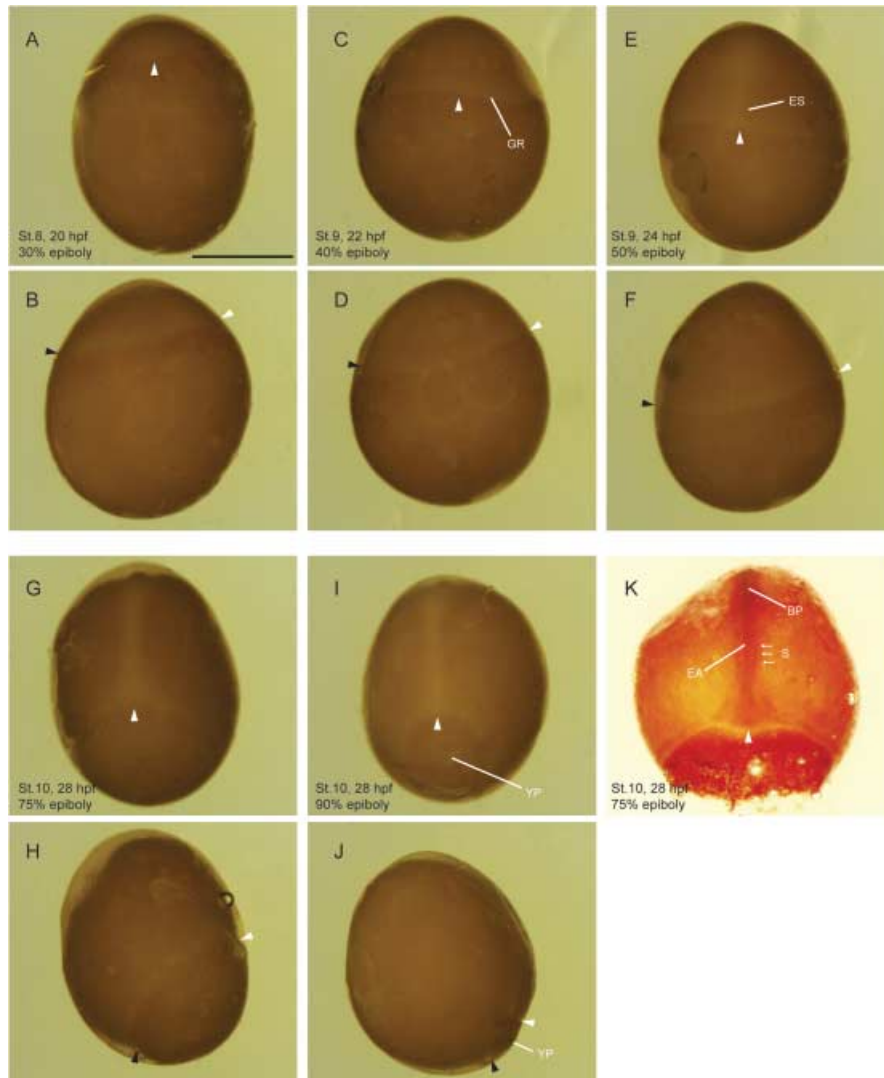


Fig. 3. Late blastula period (stage 8), gastrula period (stage 9) and early segmentation period (stage 10). Specimens were sequentially fixed with 4% paraformaldehyde in phosphate-buffered saline (PFA/PBS) and Bouin's solution. (A,C,E,G,I,K) Dorsal views of the embryo. (B,D,F,H,J) Lateral views of the embryo. From both perspectives the animal pole is at the top of the image. Scale bar, 1 mm. White arrowheads indicate the site of the blastoderm margin forming the embryonic axis. Black arrowheads indicate the opposite site of the blastoderm margin during the epibolic movements. (A,B) Stage 8 (20 hpf), 30% epiboly. The egg had cells spreading along the margin of the blastoderm. (C,D) Stage 9 (22 hpf), 40% epiboly. A thickened marginal region termed the germ ring (GR) appeared. (E,F) Stage 9 (24 hpf), 50% epiboly. Convergence movements produced a local accumulation of cells at one position along the GR, forming the embryonic shield (ES). (G,H) Stage 10 (28 hpf), 75% epiboly. The embryo became thickened, forming the embryonic axis. (I,J) Stage 10 (28 hpf), 90% epiboly. The embryo became more thickened. The bit of uncovered yolk cell was visible as a yolk plug (YP). (K) Stage 10 (28 hpf). The fixed, dechorionated embryo with 75% epiboly (panels G and H) was stained by hematoxylin. In the anterior end of the embryonic axis (EA), the primordium of the brain (BP) was visible. Three somites (S, white arrows) began to form.

in the early segmentation period from the yolk because the structures were so thin. Therefore, to distinguish them, we fixed the specimens with 4% PFA/PBS followed by treatment with Bouin's fixative, and photographed them (Figs 3A–J and 4A,E,G,K). In addition, to visualize the embryo more clearly, some of the specimens were stained with hematoxylin (Figs 3K and 4B,F,H,L). According to the zebrafish staging

system (Kimmel *et al.* 1995), we defined percent epiboly as the fraction of the yolk cell that was covered by the blastoderm.

Gastrula period (22–26 hpf)

Stage 9 (2 dpf, 22–26 hpf). Epiboly was 40% complete at 22 hpf (Fig. 3C,D). The eggs had cells spreading

along the margin of the blastoderm, filling the lateral perivitelline space. There were changes in cell density visible as more or less translucent areas in the blastoderm. The germ ring (GR in Fig. 3C), which was visible as a more translucent area, was beginning to thicken at the blastoderm margin due to involution (compare with stage 8: 30% epiboly, Fig. 3A,B). As a result of convergence and extension movements over the large yolk, epiboly began to spread the blastoderm across the yolk cell faster on the ventral side (black arrowhead in Fig. 3B,D,F,H,J) than on the dorsal side (white arrowhead in Fig. 3B,D,F,H,J).

Epiboly was 50% complete at 24 hpf (Fig. 3E,F). At this time, the rim of the blastoderm was brought midway of the egg and the axial strand became visible as the embryonic shield (ES in Fig. 3E).

Segmentation period (26–48 hpf)

Stage 10 (2 dpf, 26–30 hpf). Epiboly was between 75% and 90% complete around 28 hpf (Fig. 3G–J). The embryo became thickened, especially at the anterior end. The bit of uncovered yolk cell protruding from the region dorsal to the vegetal pole was considered a yolk plug (YP in Fig. 3I,J). In the anterior end of the embryonic axis, the primordium of the brain (BP in Fig. 3K) was visible. Somite formation (S in Fig. 3K) started in each side lateral to the forming embryonic axis (EA in Fig. 3K).

Stage 11 (2–3 dpf, 30–40 hpf). At 32 hpf, epiboly came to a close as the blastoderm completely covered the yolk plug, defining 100% epiboly (Fig. 4A). Just anterodorsal to the site of yolk plug closure, the posterior end of the embryonic axis began to form the tailbud (T in Fig. 4B,C). There were eight somites at 32 hpf (S in Fig. 4B); somites appeared sequentially in each side lateral to the forming embryonic axis (Fig. 4B). However, unlike in zebrafish (Kimmel *et al.* 1995), the number of somites did not provide a useful staging index, because it was difficult to visualize them using DIC optics in the living embryo given the relatively shallow embryo and large translucent yolk in Nile tilapia (Fig. 4C). Neurulation occurred along the forming embryonic axis (Fig. 4B) (Reichenbach *et al.* 1990). An assemblage of cells was most prominent in the anterior region of the embryo, forming the brain rudiment (Fig. 4B). The formation of the optic primordia (OpP in Fig. 4B,D) commenced with the evagination of two lateral bulges from the brain rudiment, in the anterior region of the embryo just beneath the micropyle (Balon 1977) (MP in Fig. 4D).

At 38 hpf, melanophores on the yolk surface of each side of the embryo gradually differentiated

(Fig. 4E compared with Fig. 4A). Eighteen somites were observed (S in Fig. 4F). The embryo became more thickened as the somites appeared in each side ventral to the forming neural tube in the embryo (NT in Fig. 4F). The tailbud was prominent in the posterior end of the embryo (T in Fig. 4F).

Stage 12 (3 dpf, 40–44 hpf). Melanophores became more prominent on the yolk surface of each side of the embryo than in earlier stages (white arrowheads in Fig. 4G). The cerebellum was forming (Ce in Fig. 4H,I) and the primordium of the brain began to become regionalized, forming the forebrain, midbrain and hindbrain (FB, MB and HB, respectively, in Fig. 4H). The optic primordia were observed as the optic cups and the otic placodes hollowed, forming the otic vesicle (OpP and OtV, respectively, in Fig. 4I). The otoliths were not visible at this time. On the surface of the head region, the migrating neural crest cells were visible (white asterisks in Fig. 4I) (Kimmel *et al.* 1995; Hall 1999; Le Douarin & Kalcheim 1999). The tailbud began to protrude away (Fig. 4J) and blood cells began to differentiate over the region anteroventral to the head in the perivitelline space of the yolk (see Fig. 1Ec).

Stage 13 (3 dpf, 44–48 hpf). The body of the embryo elongated during this stage (Fig. 4K,L,M). The brain thickened and visibly partitioned into the forebrain, midbrain and hindbrain (FB, MB and HB, respectively, in Fig. 4L). The ventricle of the brain was also visible (V in Fig. 4L). The olfactory placodes were prominent on the surface anterior to the optic primordia with the lens placodes (OlfP, OpP and LP, respectively, in Fig. 4M). The otic vesicle (OtV in Fig. 4M) contained two otoliths that were tiny at this time. The heart was first visible as a tube in the pericardial sac at the left ventral edge of the head. The heart began to beat at this stage. The blood cells (BC in Fig. 4M) differentiated over a broad region anteroventral to the head in the perivitelline space of the yolk (Fig. 1Ed and Fig. 4M). However, the blood had not begun to circulate.

Developmental stages in the pharyngula and hatching periods of *O. niloticus*

In this section, we describe stages 14–18 in the pharyngula and hatching periods. Craniofacial development of *O. niloticus*, especially differentiation and morphogenesis of the pharyngeal skeleton, progressed in the pharyngula and hatching periods. According to the zebrafish staging system (Kimmel *et al.* 1995), we defined two characteristics: embryo length (EL) referred to the embryo's longest linear dimension along the anteroposterior axis from stages 14–18,

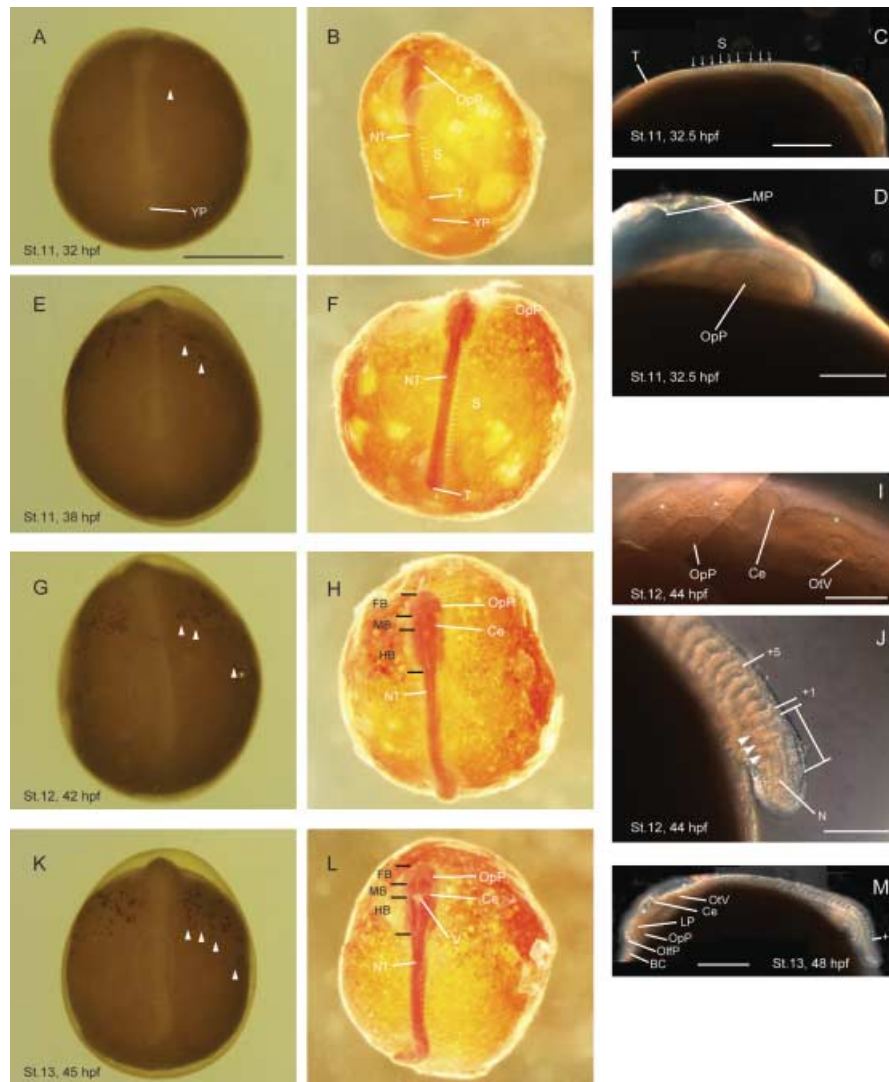


Fig. 4. Segmentation period (stages 11–13). (A,E,G,K) Dorsal view of the embryo fixed with 4% paraformaldehyde in phosphate-buffered saline (PFA/PBS) and refixed with Bouin's fixative. White arrowheads indicate the melanophores on the egg surface of each side of the embryo. (B,F,H,L) The embryos were dechorionated and stained with hematoxylin. These images are shown with the animal pole up. Scale bar, 1 mm. (C,D,I,J,M) Views of the living embryos through Nomarski differential interference contrast (DIC) optics. Scale bar, 500 μ m (C and M), 200 μ m (D, I, J). Ce, cerebellum; FB, forebrain; HB, hindbrain; LP, lens placodes; MB, midbrain; N, notochord; OlP, olfactory placodes; OpP, optic primordium; OtV, otic vesicles; T, tailbud; V, ventricle of the brain. (A,B) Stage 11 (32 hpf). The yolk plug (YP) was closed. Eight somites (S, white arrows) formed. Somites appeared sequentially in each side lateral to the forming neural tube (NT) on the yolk. (C) Stage 11 (32.5 hpf). Anterior is at right. Nine somites (S, white arrows) formed. (D) Stage 11 (32.5 hpf). Anterior is at right. The OpP were observed as the evaginations of two lateral bulges from the brain primordium in the anterior region of the embryo, just beneath the micropyle (MP). (E,F) Stage 11 (38 hpf). Eighteen somites (S, white arrows) formed. (G,H) Stage 12 (42 hpf). Because the somites were ventral to the embryo, they were not visible from the dorsal view. The primordium of brain began to regionalize. (I) Stage 12 (44 hpf). On the surface of the head region, the migrating neural crest cells (white asterisks) were visible. (J) Stage 12 (44 hpf). The tail was undercut. In the differentiating tissue (section indicated by white lines) of the tail region, the rudiments of three somites (white arrowheads) were forming. '+1' Indicates the most posterior somite and we could count up the somites in the anterior direction (e.g. '+5'). (K,L) Stage 13 (45 hpf). The brain was apparently partitioned. (M) Stage 13 (48 hpf). On the surface of the head region, the migrating neural crest cells (white asterisks) were visible. The brain thickened. The tail was elongated. '+1' Indicates the most posterior somite. The blood cells (BC) over the perivitelline space were visible anteroventral to the head.

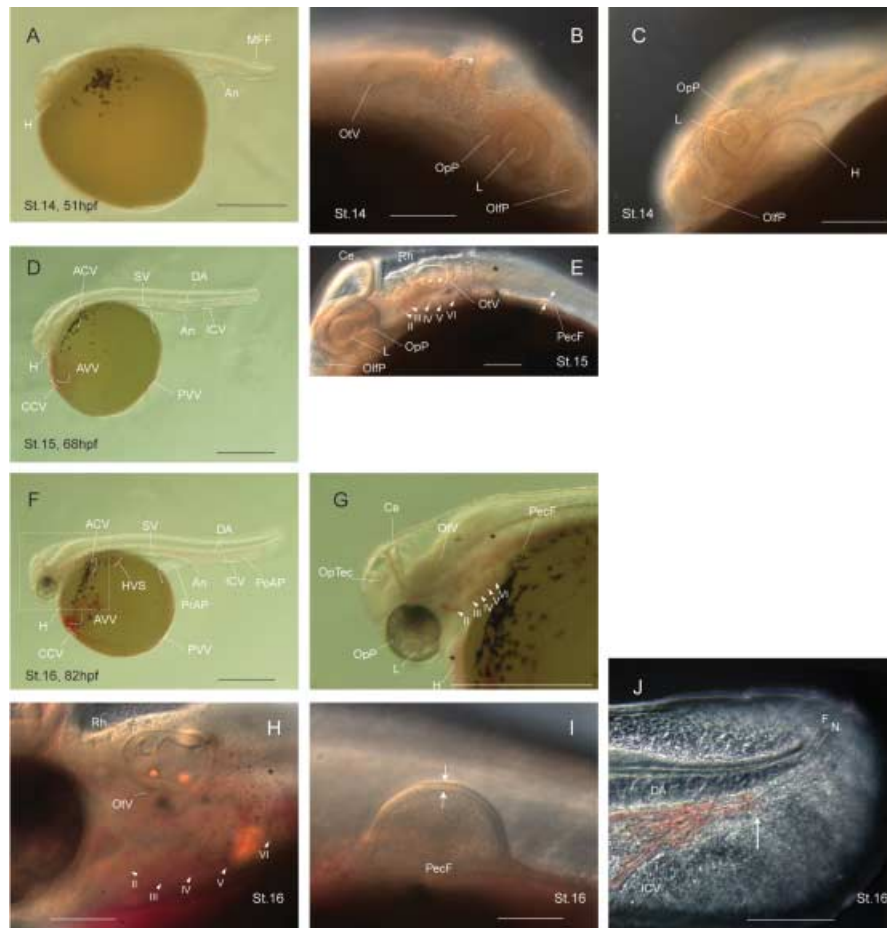


Fig. 5. Pharyngula period (stages 14–16). (A,D,F,G) Lateral views of the living and dechorionated embryos. Magnification of the area boxed in panel F is shown in panel G. Scale bar, 1 mm. (B,C,E,H–J) DIC views of the living and dechorionated embryos. Scale bar, 200 μ m, except for panel I (100 μ m). ACV, anterior cardinal vein; AVV, anterior vitelline vein; CCV, common cardinal vein; Ce, cerebellum; DA, dorsal aorta; H, heart; HVS, hepatic vitelline system; ICV, inferior caudal vein; L, lens; OlfP, olfactory placodes; OpP, optic primordium; OpTec, optic tectum; OtV, otic vesicles; PoAP, postanal plexus; PrAP, preanal plexus; PVV, posterior vitelline vein; Rh, rhombomeres; SV, subintestinal vein; II, aortic arch 2 (AA2) in the hyoid arch; III, AA3 in the first branchial arch; IV, AA4 in the second branchial arch; V, AA5 in the third branchial arch; VI, AA6 in the fourth branchial arch. (A) Stage 14 (51 hpf). The median fin fold (MFF) began to form in the tail region. The anus (An) began to form. (B,C) Stage 14 (53 hpf). (B) View of the right side of the head and (C) view of the left side. Migrated cranial neural crest cells (white asterisk) were observed on the surface of the head. (D) Stage 15 (68 hpf). The blood circulation started. (E) Stage 15 (68 hpf). The brain was sharply differentiated. Five pairs of aortic arches (AA) began to form (white arrowheads). The arrows that face each other indicate the apical ectodermal ridge (AER) of the pectoral fin buds (PecF). The hatching glands (black asterisk) were present on the surface of each side of the embryo anterodorsal to pectoral fin buds. (F,G) Stage 16 (82 hpf). The head and pharyngeal region with the AA (white arrowheads) were more enlarged. The hatching glands (black asterisks) were present on the surface of the pericardial cavity as well as on each side of the embryo. (H) Stage 16 (80 hpf). The primordia of the semicircular canals began to form dorsally in the otic vesicle (OtV). Numerous hatching glands (black asterisk) were present. (I) Stage 16 (78 hpf). The arrows that face each other indicate the AER of the pectoral fin buds. (J) Stage 16 (80 hpf). Circulation from the DA to the ICV partially supplied blood (arrows) to the tail fin beneath the floor plate (F) of the spinal cord and the notochord (N).

and otic vesicle length (OVL) referred to the estimated number of additional otic vesicles that could fit between the eye and the otic vesicle in stages 14–17.

Pharyngula period (48–90 hpf)

Stage 14 (3–4 dpf, 48–60 hpf), 50 hpf: EL = 3.3 mm, OVL = 4. The tail region straightened out (Fig. 5A)

and the posterior end of the trunk-tail was undercut and elongated. There were about 32 somites, 14 of them in the tail. The median fin fold began to form in the tail region and the anus began to form (MFF and An, respectively, in Fig. 5A). The optic cups with lens placodes were discernible (Figs 5B,C). The otic vesicles were lateral to rhombomeres 3 and 4 as the wall thickened. The cells that occupied the

surface region between the optic cup and the otic vesicle seemed to be from cranial neural crest cells and began to differentiate (white asterisk in Fig. 5B). The olfactory placodes were prominent (OlfP in Fig. 5B,C). The heart began to beat as a waving tube (H in Fig. 5C). The primordial dorsal aorta began to develop ventral to the notochord, but appeared empty.

Stage 15 (4 dpf, 60–72 hpf), 67 hpf: EL = 4.0 ± 0.2 mm, OVL = 1–1.5. The brain was sharply differentiated and the cerebellum and rhombomeres (Ce and Rh, respectively, in Fig. 5E) were apparent. There were ovoid otic vesicles (OtV in Fig. 5E) as thickening walls with the two otoliths. The pectoral fins (PecF in Fig. 5E) were budding on the yolk surface near the middle region of the trunk parallel to the body axis. Like zebrafish (Fig. 37 in Kimmel *et al.* 1995), the apical ectodermal ridge (AER) was present at the tip of the pectoral fin bud and was easily visible with DIC optics (arrows in Fig. 5E). The height-to-width ratio of the pectoral fins was less than 1:4. The hatching glands (black asterisk in Fig. 5E) were present in each side of the embryo on the surface anterodorsal to the pectoral fin buds. Blood circulation started as the heart differentiated with expansion of the pericardial cavity. Blood from the heart (H in Fig. 5D) circulated through the aortic arches and then circulated either around the head or the trunk-tail regions. The five pairs of aortic arches (AAII, AAIII, AAIV, AAV, AAVI) began to form (II, III, IV, V, VI, respectively, in Fig. 5E) and it appeared that the pharyngeal arches had formed. The blood that circulated around the head region was collected into the anterior cardinal vein (ACV in Fig. 5D). The blood of the anterior cardinal vein entered the anterior vitelline vein (AVV in Fig. 5D) on the surface of the yolk. The anterior vitelline vein passed just anterior to the pectoral fin bud. The blood that circulated around the trunk-tail region through the dorsal aorta (DA in Fig. 5D) was collected in the subintestinal vein or the inferior caudal vein (SV and ICV, respectively, in Fig. 5D). The postanal plexus of the inferior caudal vein began to form in the postanal fin fold, which was the ventral part of the median fin fold. The blood entered the yolk via numerous branches in the subintestinal and preanal fin fold. The blood from the veins entered the posterior vitelline vein (PVV in Fig. 5D) on the surface of the yolk. Blood from the anterior vitelline vein and the posterior vitelline vein entered the common cardinal vein (CVV in Fig. 5D) and then entered the heart tube. There were melanophores on the ventral side of the embryo along the anterior vitelline vein.

Stage 16 (4–5 dpf, 72–90 hpf), 72 hpf: EL = 4.3 ± 0.1 mm, OVL = 0.8–1. The head and the pharyngeal region with the aortic arches were more enlarged. The optic tectum (OpTec in Fig. 5G) was differentiated in the midbrain. The primordia of the semicircular canals began to form dorsally in the otic vesicle (OtV in Fig. 5H) and the pectoral fin buds were elongated with the AER (arrows in Fig. 5I). The height-to-width ratio of the pectoral fin buds was about 1:2. The tip of the notochord began to bend gradually upward (N in Fig. 5J). Numerous hatching glands were present on the surface dorsal to the pectoral fins and of the pericardial sac (black asterisks in Fig. 5G,H). The postanal plexus of the inferior caudal vein (PoAP in Fig. 5F) and the preanal plexus of the inferior caudal vein and the subintestinal vein (PrAP in Fig. 5F) formed. Circulation from the dorsal aorta to the inferior caudal vein was supplied partially in the tail, beneath the floor plate of the spinal cord and the notochord (DA, ICV, F and N in Fig. 5J). Posterior to the anterior cardinal vein, the veins of the hepatic vitelline system covered the surface of the yolk (ACV and HVS in Fig. 5F). More melanophores appeared in the eyes around 70 hpf and one could easily distinguish the eyes of embryos in this stage (Fig. 5F,G).

Hatching period (90–120 hpf)

Stage 17 (5–6 dpf, 90–110 hpf), 90 hpf: EL = 4.9 ± 0.1 mm, OVL = 0.5. Prominent changes occurred in the pharyngeal region. The mouth protruded as the lower jaw (black arrowheads in Fig. 6B,D,F,H) gradually elongated and the head straightened out (white arrows in Fig. 6B,D,F,H). The mouth was closed by a thin epithelial sheet, the oropharyngeal membrane (Fig. 6B,D), but began to open (Fig. 6F,H). The pectoral fin rudiments (PecF in Fig. 6B,D,F,H) were elongated buds already containing centrally located mesenchymal condensations that would form the girdle cartilages. The height-to-width ratio of the pectoral fin rudiments was 1:1. There was yet no blood circulation in the pectoral fin buds. Embryos in this stage usually hatched. The heart gradually moved into position under the head (Fig. 6B,D,F,H). The bend in the heart was prominent and marked the division between the atrium and the ventricle. Each gill arch (III, IV, V, VI in Fig. 6B,D,F,H) began to differentiate. The opercular artery appeared posterior to the artery of AAII. These AAII and opercular arteries formed an inverted U-shape on the surface posteroventral to the eye. In the caudal fin, the blood circulation from the dorsal aorta to the inferior caudal vein expanded to form the urostylar artery and vein and a loop in the hypural

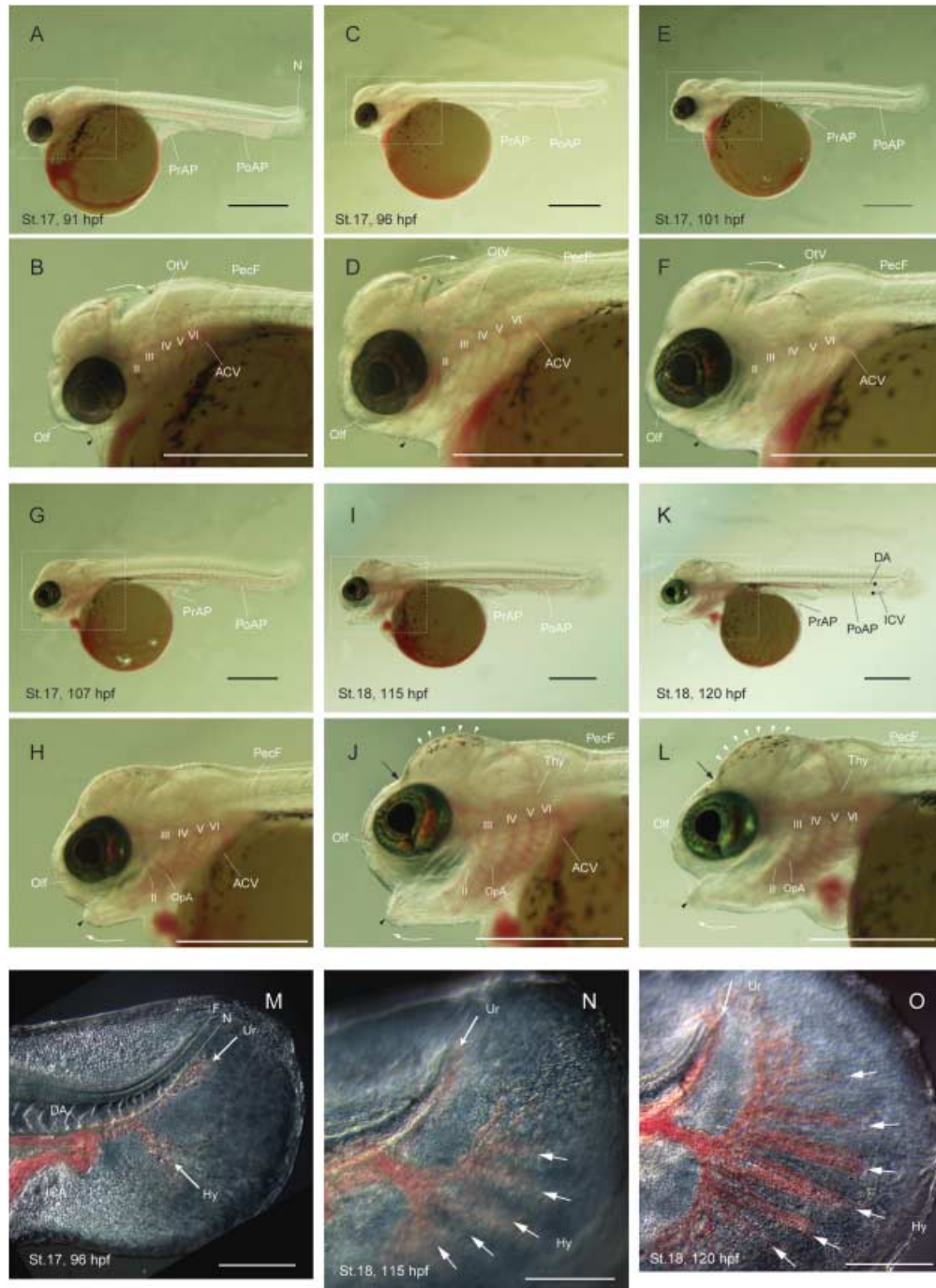


Fig. 6. Hatching period (stages 17–18). (A,C,E,G,I,K) Lateral views of the living embryos. Scale bar, 1 mm. (B,D,F,H,J,L) Magnification of the head area boxed in panels A, C, E, G, I, K, respectively. Scale bar, 1 mm. (M,N,O) DIC views of the caudal fin. Scale bar, 200 μ m. ACV, anterior cardinal vein; DA, dorsal aorta; N, notochord; Olf, olfactory organ; OpA, opercular artery; OtV, otic vesicle; PecF, pectoral fin; PoAP, postanal plexus; PrAP, preanal plexus; Thy, thymus glands. II, aortic arch 2 (AA2) in the hyoid arch; III, AA3 in the first branchial arch; IV, AA4 in the second branchial arch; V, AA5 in the third branchial arch; VI, AA6 in the fourth branchial arch. (A,B) Stage 17 (91 hpf); (C,D) stage 17 (96 hpf); (E,F) stage 17 (101 hpf); (G,H) stage 17 (107 hpf). The mouth protruded as the lower jaw (black arrowheads) gradually elongated and the head straightened out (white arrows). (I,J) Stage 18 (115 hpf); (K,L) stage 18 (120 hpf). Melanophores gradually appeared on the surface of the head (white arrowheads). As the midbrain protruded dorsally, it appeared from the lateral view that the position at the epiphyseal bar (black arrow) became depressed. (M) Stage 17 (96 hpf); (N) stage 18 (115 hpf); (O) stage 18 (120 hpf). Ventral to the floor plate of the spinal cord (F) and the notochord (N) in the caudal fin, the blood circulation (white arrows) from the DA to the ICA expanded to form the urostyle artery and vein (Ur) and loops in the hypural region (Hy).

region (DA, Ur and Hy, respectively, in Fig. 6M). Some of the intersegmental arteries and veins began to form. Pigmentation of the eye was very dense and some iridophores in the eyes appeared (Fig. 6B,D,F,H). A wide band of melanophores began to cover the upper part of the gut (Fig. 6A,C,E,G).

Stage 18 (6 dpf, 110–120 hpf), 114 hpf: EL = 5.8 ± 0.2 mm. Each gill arch (III, IV, V, VI) had developed a cartilaginous rod and gill filaments formed (Fig. 6J,L). The gill cover and branchiostegal membrane were forming. The thymus (Thy in Fig. 6L) formed as a bulge on the surface dorsolateral to the pharyngeal cavity (Fig. 6J,L) (Fishelson 1995b). As the midbrain protruded dorsally, it appeared from the lateral view that the position at the epiphyseal bar became depressed (black arrow in Fig. 6J,L). In each pectoral fin, a blood vessel was present in the middle of the proximodistal axis. In the hypural region of the caudal fin, the blood circulation formed more loops and the caudal fin rays began to form (Hy in Fig. 6N,O). The postanal plexus began to decline (PoAP in Fig. 6I,K). The preanal plexus was also markedly reduced (PrAP in Fig. 6I,K). There were numerous melanophores in the gut cavity and on the dorsal surface of the head (Fig. 6I,K, white arrowheads in Fig. 6J,L). Also, melanophores extended to the end of the notochord along the dorsal aorta and the inferior caudal vein. More iridophores appeared in the eyes (Fig. 6J,L).

Developmental stages in the early larval period of *O. niloticus*

In this section, we describe stages 19–22 in the early larval period. The early larvae gradually began to move its jaws, opercular flaps and pectoral fins.

Following Morrison *et al.* (2001), we defined larval length (LL) as the larva's longest linear dimension along the anteroposterior axis from stages 19–22. Moreover, according to Otten (1982), we identified the number of CFRE in stages 20–32 (Fig. 2).

Early larval period

Stage 19 (6–7 dpf, 120–132 hpf), LL = 6.2 ± 0.1 mm. Movements of the gills and the mouth were much stronger (Fig. 7A,B). The gills were still external and were not completely covered by the operculum (Fig. 7B). The fin fold in the area of the dorsal and anal fins began to differentiate, and at the base of each of these fins an aggregation of mesenchyme could be seen (DF and AF in Fig. 7C). In the ventral part of the tail region, the inferior caudal vein (ICV

in Fig. 7C) was in the process of being replaced by the profundal caudal vein (PCV in Fig. 7C). The large yolk was covered with capillaries. There were more melanophores in the trunk-tail and head regions than in earlier stages (Fig. 7A–C) and some melanophores beneath the pharynx (white asterisk in Fig. 7B).

Stage 20 (7 dpf, 132–144 hpf), LL = 6.5 ± 0.2 mm. The number of CFRE was two; thus, the rays of the caudal fin formed into two segments. The inflation of swimbladder started, but it did not make it possible for the larva to float completely because the yolk was still large. There were more aggregated cells in the area of dorsal and anal fins than in the previous stage, and the dorsal and anal fins began to be distinguishable (DF and AF in Fig. 7F). The pseudobranch (PB in Fig. 7E) that was circulated by the AA2 began to form (Shadrin & Ozernyuk 2002). In the ventral part of the tail region, the inferior caudal vein was replaced by the profundal caudal vein (PCV in Fig. 7F). A pair of teeth was visible protruding from the mandible (white arrowhead Fig. 7M) (Huysseune & Sire 1997; Streelman *et al.* 2003; Streelman & Albertson 2006). Gills were almost completely covered by the operculum (Fig. 7E). Some of the branchiostegal rays began to form in the branchiostegal membrane (BR in Fig. 7E). More melanophores were visible at the top of the head and beneath the pharynx (Fig. 7E). An intensely yellow mass began to fill the gut (black asterisk in Fig. 7F).

Stage 21 (7–8 dpf), LL = 6.9 ± 0.2 mm, CFRE = 3. The number of CFRE increased quickly from two to three. The upper jaw bones and the cleithrum formed (PMa, Ma, Cl in Fig. 7H). There were more aggregated cells in the area of dorsal and anal fins than in the previous stage (Fig. 7I). More melanophores were visible at the top of the head, beneath the pharynx and the trunk-tail regions (Fig. 7H,I). Some iridophores began to differentiate on the surface lateral to the posterior of the gut (Fig. 7G). The heart was gradually put into the space beneath the pharynx (H in Fig. 7H).

Stage 22 (8–9 dpf), LL = 7.2 ± 0.3 mm, CFRE = 4. Yolk absorption started and the size of the yolk was smaller than in the previous stages, as the body wall with the tissue that Holden and Bruton (1992) called 'unidentified white tissue' descended (white broken line in Fig. 7J and UWT in Fig. 7O). In the 'unidentified white tissue', the pelvic fin buds were present on the surface of the yolk and one could observe them from the dorsal view (PelF in Fig. 7O). Unlike the pectoral fin buds (arrows in Fig. 5E,I), the AER of the pelvic

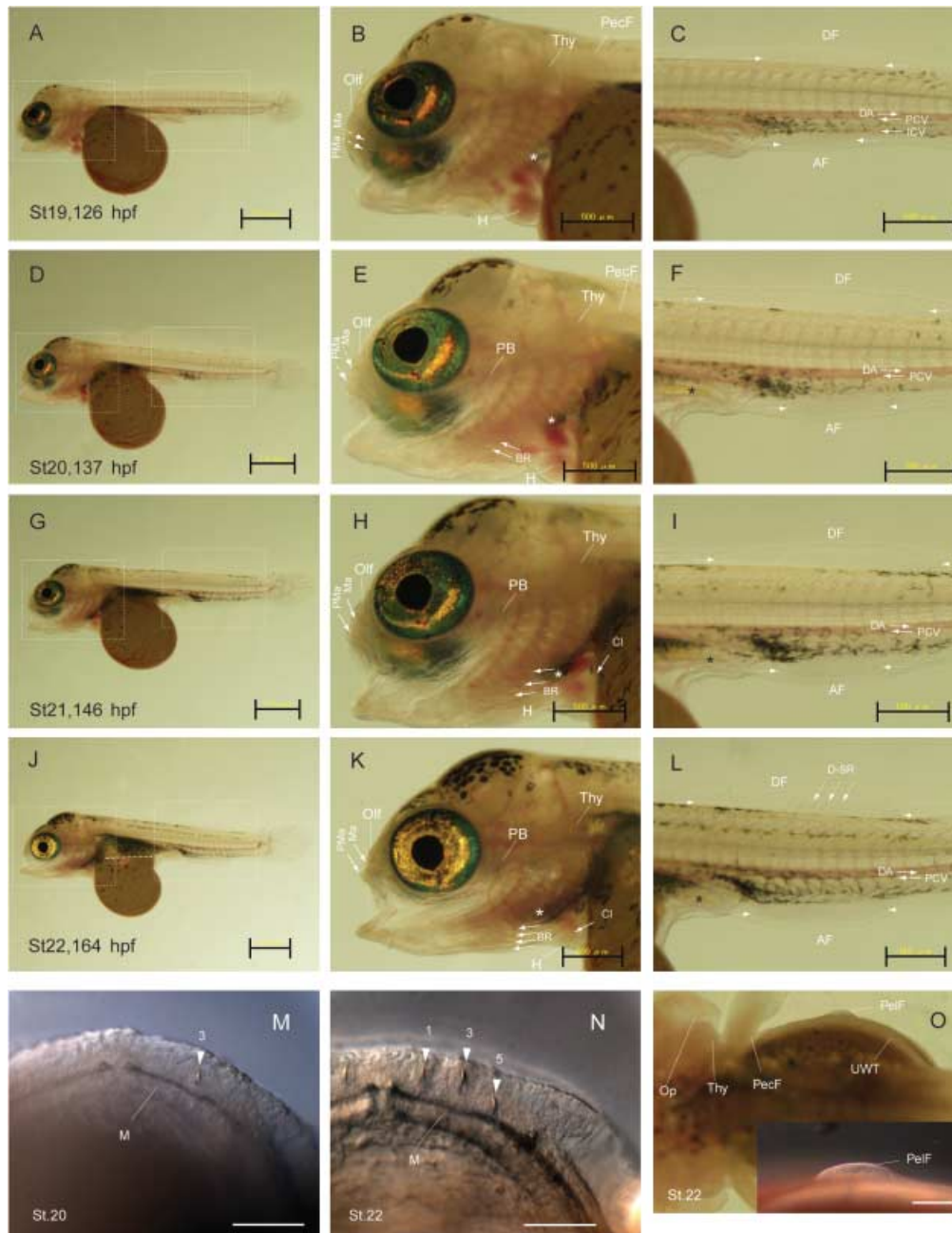


Fig. 7. Early larval period (stages 19–22). (A,D,G,J) Lateral views of the living larvae. Scale bar, 1 mm. (B,E,H,K) Magnification of the head area boxed in panels A, D, G, J, respectively. Scale bar, 500 μ m. (C,F,I,L) Magnification of the middle area boxed in panels A, D, G, J, respectively. Scale bar, 500 μ m. (M, N, inset of O) DIC views of the living larvae. Scale bar, 200 μ m, except for panel O (100 μ m). AF, anal fin; BR, branchiostegal rays; Cl, cleithrum; DA, dorsal aorta; DF, dorsal fin; D-SR, dorsal soft rays; H, heart; ICV, inferior caudal fin; Ma, maxilla; Olf, olfactory organ; Op, operculum; PB, pseudobranch; PCV, profundal caudal vein; PecF, pectoral fin; PMa, premaxilla; Thy, thymus gland. (A–C) Stage 19 (126 hpf). The gills were still external and not completely covered by the operculum. The fin fold in the area of the dorsal and anal fins began to differentiate and the broken arrows indicate aggregations of mesenchyme. (D–F) Stage 20 (137 hpf). Gills were almost completely covered by the operculum. Black asterisk indicates that an intensely yellow mass began to fill the gut. (G–I) Stage 21 (146 hpf). The upper jaw bones formed (PMa and Ma). (J–L) Stage 22 (164 hpf). Yolk absorption started and the body wall descended (white broken line). The broken arrows indicate that the dorsal soft rays in the posterior part of the dorsal fin began to form. (M) Stage 20 (144 hpf). View of the dorsal side of the mandible. A pair of teeth developed, protruding from the mandible (M, Meckel's cartilage) at position 3. (N) Stage 22 (168 hpf). View of the ventral side of the mandible. Three pairs of teeth developed at positions 1, 3 and 5. (O) Stage 22 (175 hpf). From the dorsal view, the pelvic fin buds (PelF) were observed in the undifferentiated white tissue (UWT) which formed the body wall with absorption of the yolk. (Inset of panel O) An apical ectodermal ridge-like structure was not observed in the pelvic fin bud (PelF).

fin bud was not observed (Fig. 7O). The dorsal soft rays in the posterior part of the dorsal fin began to form (broken arrows of DF in Fig. 7L). Three pairs of teeth were observed on the mandible (white arrowheads in Fig. 7N). More melanophores were visible at the top of the head and beneath the pharynx (Fig. 7K). There were numerous melanophores in the trunk-tail regions (Fig. 7L). There were numerous iridophores on the surface lateral to the gut (Fig. 7J). There were more iridophores on the surface of the operculum than in the previous stage (Fig. 7K).

Developmental stages in the late larval and early juvenile periods of *O. niloticus*

In this section, we describe stages 23–32 in the late larval and early juvenile periods. Craniofacial morphologies were functionalized during these periods. We defined total length (TL) as the longest linear dimension between the anterior tip of the upper jaw (premaxilla) and the posterior end of the caudal fin.

Late larval period

Stage 23 (9–10 dpf), TL = 8.1 ± 0.2 mm, CFRE = 5. Inflation of the swim bladder allowed the fish to float (SB in Fig. 8A). The dorsal soft rays in the posterior part of dorsal fin were forming (D-SR in Fig. 8C), while an aggregation of mesenchyme was still in the anterior part of the dorsal fin and in the anal fin (broken arrows in Fig. 8C). The fin fold behind the dorsal and anal fins began to be gradually absorbed and thus separated the caudal fin to form the caudal peduncle (black arrows in Fig. 8C). There were numerous melanophores on the surface of the trunk-tail regions (Fig. 8C). There were more iridophores on the surface of operculum than in the previous stage (Fig. 8B). There was also ‘unidentified white tissue’ with numerous iridophores covering 50% of the yolk as a result of yolk absorption (white broken line in Fig. 8A). The pelvic fin buds were on the tip of the cover. Five or six pairs of teeth were observed on the mandible (white arrowheads in Fig. 8J). The order of teeth development is 3–5/1–2/4–7 (cf. rainbow trout, Berkovitz 1977). Teeth numbered 1, 3 and 5 were anchored into the mandible (Huysseune & Sire 1997).

Stage 24 (9–10 dpf), TL = 8.6 ± 0.2 mm, CFRE = 6. As the yolk was substantially absorbed, 75% appeared to be covered by ‘unidentified white tissue’ (white broken line in Fig. 8D). The swimbladder was fully developed (SB in Fig. 8D). Dorsal spines in the anterior part of the dorsal fin and anal soft rays in the anal

fin began to form (broken arrows of D-Sp and A-SR in Fig. 8F). The pelvic fin buds were observed from the ventral view (PelF in Fig. 8K). The fin buds looked like an aggregation of undifferentiated cells (inset of Fig. 8K). Morphogenesis of the lateral line canals along the posteroventral part of the lower jaws commenced (Munz 1979; Tarby & Webb 2003) and the neuromasts (broken lines and NM in Fig. 8L) were observed (Webb 1989, 1990). There were more melanophores on the surface of the trunk-tail and head regions than in the previous stage (Fig. 8D,E,F). There were more iridophores on the surface of the operculum than in the previous stage (Fig. 8E).

Stage 25 (11–13 dpf), TL = 9.0 ± 0.4 mm, CFRE = 7. As the yolk absorption progressed further, more than 90% of the yolk seemed to be covered (Fig. 8G). The fish began foraging for food (e.g. brine shrimp). The pelvic fin buds were observed from the lateral view (PelF in Fig. 8G). The dorsal spines and the dorsal soft rays in the dorsal fin and the anal soft rays in the anal fin were forming (Fig. 8I). Anal spines began to form in the anterior part of the anal fin (broken arrows in Fig. 8I). There were more melanophores on the surface of the trunk-tail and head regions than in the previous stage (Figs 8G,H,I). There were more iridophores on the surface of the operculum than in the previous stage (Fig. 8H). Melanophores for the tilapia mark began to form in the middle of the dorsal fin (TM in Fig. 8I).

Early juvenile period

Stage 26 (12–13 dpf), TL = 9.9 ± 0.4 mm, CFRE = 8. As the yolk was completely absorbed within the body cavity walls (although remnants of it remained under the intestine), it appeared to be 100% covered (Fig. 9A). There were melanophores dispersed over the dorsal fin (Fig. 9B). Melanophores for the tilapia mark were more distinguishable than in the previous stage, although they were sparse (TM in Fig. 9B). Rays of the pelvic fin began to differentiate and an aggregation of mesenchyme could be seen in the pelvic fin (broken arrows in Fig. 9C).

Stage 27 (14–15 dpf), TL = 10.4 ± 0.4 mm, CFRE = 9. There were melanophores at the anterodorsal margin of the dorsal fin (Fig. 9D,E). The tilapia mark was more distinct (TM in Fig. 9E). Rays of the pelvic fins differentiated (arrows in Fig. 9F).

Stages 28–32, TL = 11.9 ± 0.4 mm – 17.3 ± 0.8 mm, CFRE = 10–14. The vertical bands of melanophores and iridophores gradually formed on the sides of



Fig. 8. Late larval period (stages 23–25). (A,D,G) Lateral views of the living larvae. Scale bar, 1 mm. (B,E,H) Magnification of the head area boxed in panels A, D, G, respectively. Scale bar, 500 μ m. (C,F,I) Magnification of the middle area boxed in panels A, D, G, respectively. Black arrows indicate that the fin fold behind the dorsal and anal fins began to be gradually absorbed. Scale bar, 500 μ m. (K) Ventral view of the living larvae. (J, inset of K, L) DIC views of the living larvae. Scale bar, 100 μ m. A-Sp, anal spines; A-SR, anal soft rays; D-Sp, dorsal spines; D-SR, dorsal soft rays. Ma, maxilla; Olf, olfactory organ; PelF, pelvic fin; PMa, premaxilla; SB, swimbladder; UWT, undifferentiated white tissue. (A–C) Stage 23. The body wall covered 50% of the yolk and contained 'unidentified white tissue' with numerous iridophores. The dorsal soft rays (D-SR) in the posterior part of dorsal fin were forming. (D–F) Stage 24. The body wall covered 75% of the yolk. Broken arrows indicate that the dorsal spines (D-Sp) in the anterior part of dorsal fin and the anal soft rays (A-SR) in the anal fin began to form. (G–I) Stage 25. The body wall covered more than 90% of the yolk. Broken arrows indicate that anal spines (A-Sp) began to form in the anterior part of anal fin. Melanophores for the tilapia mark (TM) began to form in the middle of the dorsal fin. (J) Stage 23. View of the ventral side of the mandible. Arrowheads indicate that five teeth were observed on one side of the mandible (M, Meckel's cartilage) and six teeth were on the opposite side. (K) Stage 24. From the ventral view, the pelvic fin buds (PelF) were visible in the lateral side of the descending body wall. (Inset of panel K) The pelvic fin buds looked like an aggregation of undifferentiated cells. (L) Stage 24. View of the lateral side of the mandible. Anterior is at left. The white broken line indicates that morphogenesis of the lateral line canals at the lower jaws began. Neuromasts (NM) were observed on the posteroventral surface of the mandible.

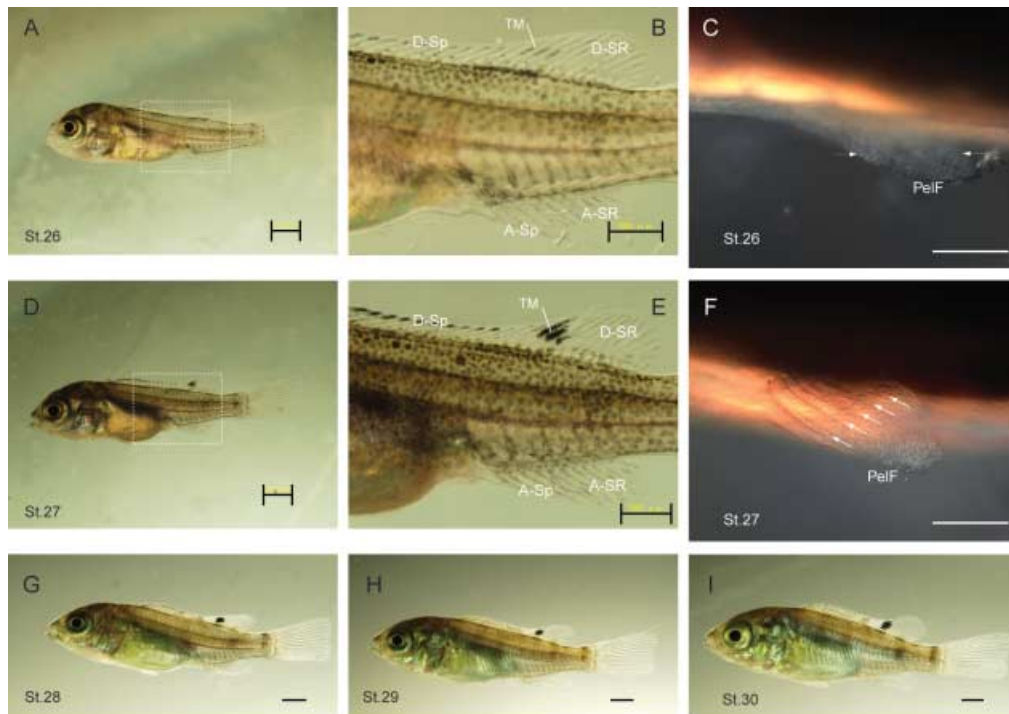


Fig. 9. Early juvenile period (stages 26–32). (A,D,G–I) Lateral views of the living juveniles. Scale bar, 1 mm. (B,E) Magnification of the area boxed in panels A and D, respectively. Scale bar, 500 μ m. (C,F) DIC views of the lateral side of the pelvic fin. Scale bar, 200 μ m. A-Sp, anal spines; A-SR, anal soft rays; D-Sp, dorsal spines; D-SR, dorsal soft rays. (A,B) Stage 26. The yolk was completely absorbed and covered by the body wall. In the middle of the dorsal fin, melanophores for the tilapia mark (TM) were more distinct than in the previous stage. (C) Stage 26. Broken arrows indicate that rays of the pelvic fin began to differentiate. (D,E) Stage 27. There were melanophores at the anterodorsal margin of the dorsal fin as well as in the TM. (F) Stage 27. Arrows indicate that the rays of the pelvic fins were differentiated. (G) Stage 28; (H) stage 29; (I) stage 30. Vertical bands of melanophores and iridophores gradually formed on the sides.

juveniles (Fig. 9G–I). Except for the increases in TL and CFRE, no apparent changes in the external characteristics were observed between stages 28 and 32 (Table 1, Fig. 1J,c–g, Fig. 9G–I).

Discussion

Description of the developmental stages in *O. niloticus*

There are four notable points regarding our description of the developmental stages observed in *O. niloticus*: (1) we comprehensively defined the stages of cichlid development for 30 dpf. Craniofacial development of cichlids, in particular, differentiation and morphogenesis of the pharyngeal skeleton, progressed until about 30 dpf. (2) We used numbered stages rather than named stages, which made it easier to remember and recognize the sequence of development. The

numbered staging system in the present study was modified from two different staging systems that were previously described. Stages 1–19 were based on external characteristics (Galman & Avtalion 1989) and stages 20–32 were defined using the number of CFRE (Otten 1982).

Length records (EL or TL/standard length (SL)) can be considered an indication of fish development. An increase in the number of CFRE was accompanied by an increase in the body length and the number of CFRE could easily be used as an index for the length records. The staging system of Otten (1982) has been used to define the development in other mouth-brooder cichlids (e.g. *Astatotilapia burtoni*, Huysseune 1990). Therefore, we suggest that the number of CFRE is a good indicator, not only to define the later developmental stages in *O. niloticus*, but also for easy stage assignment among all other mouth-brooder cichlids (see below).

(3) We also referred to the staging system of Morrison *et al.* (2001) with some modification, from which we used developmental intervals as general terminology. We used these periods to group the numbered stage(s) into larger time blocks. Of the periods, we used the term 'larva' for the period between hatching and the end of yolk absorption. Balon (1975) emphasized that mouth-brooder cichlids do not have a larval period, because metamorphosis has not occurred and the development is direct. The term 'larva' is often used for the period of fish with metamorphosis and altricial development, and the fish in the larval stage bear little resemblance to the sexually mature fish (Kunz 2004). However, during ontogeny, all fish species experience a period of postembryonic change in structure, with the degree of change varying between species (Youson 1988; Sanderson & Kupferberg 1999). This fact has led to extensive and ongoing debate regarding the terminology (Balon 1975, 1990; Kendall *et al.* 1984; Kunz 2004).

(4) The stages were based on morphological features, identified by examining embryos, larvae and juveniles with Nomarski DIC optics as well as a dissecting stereomicroscope.

Expandability of this developmental staging system

In the present study, we defined 32 stages in the development of *O. niloticus*. This number of stages, however, might not be sufficient to elucidate more detailed sequential craniofacial development events. In such cases, this system could be divided into more detailed substages. For example, at stage 17, several craniofacial morphological changes occurred (Fig. 6A–H), including jaw elongation and chondrogenesis. Thus, this stage could be further divided to describe the precise details of craniofacial development.

Most studies of fish development adopt time records (hpf or dpf) to standardize the developmental sequences, in some cases, in combination with length records (EL or TL/SL). One can gain more useful information by using the staging system of this study that incorporates several data (e.g. stage 17, 96 hpf), because experiments based on recording with reference to a staging series such as ours provide a good way to ensure reproducibility and to allow subsequent incorporation of new observations (Kimmel *et al.* 1995).

Tilapia in evolutionary developmental biology of fish

In Figure 10, we compared the development of Nile tilapia (a mouth-brooder cichlid) and that of other fresh-water fish (two model fish, zebrafish and medaka, and a substrate-brooder cichlid *Cichlasoma*

dimerus) as far as we could. Cichlids belong to order Perciformes and represent higher teleosts. Therefore, the comparative development studies using model fish and Nile tilapia will play an important role in fish evo-devo.

Many of the marine fish have distinctive larval stages and undergo metamorphosis. It was reported that metamorphosis in several teleost species is mediated by thyroid hormones (Youson 1988). Although zebrafish does not undergo dramatic metamorphosis, Brown (1997) and Liu and Chan (2002) suggested that thyroid hormones play a substantial role in the two transitions 'from embryonic to larval' and 'from larval to juvenile' in zebrafish (T1 and T2 of zebrafish in Fig. 10), respectively. We presumed the transitions of the other fish by the morphological changes (T1 and T2 of the other fish in Fig. 10) described by Liu and Chan (2002): 'protrusion of mouth' (M in Fig. 10) for T1; 'differentiation of unpaired fins' (UPF in Fig. 10) and 'differentiation of pectoral fins' (PecF in Fig. 10) for T2. In the earlier development than T1, the development of each fish seems to share the same characteristics along to the developmental course; formations of the embryonic shield, the cerebellar primordium, the pectoral fin bud, and the aortic arches (Fig. 10). Conversely, in the later development than T1, the development of Nile tilapia and that of other fish are apparently different; T2 of Nile tilapia occurred before yolk absorption, while T2 of the other fish occurred after yolk absorption.

Several dynamic changes occurred in the early larval period of Nile tilapia: (1) differentiation of unpaired fin; (2) development of teeth; (3) differentiation of several dermal bones; (4) formation of pelvic fin buds; (5) inflation of swimbladder; (6) appearance of iridophores; (7) expansion of melanophores; and (8) maturation of intestine. These events are considered 'larval to juvenile' transition of Nile tilapia. The transition of Nile tilapia may be also mediated by some endocrinal stimulation including thyroid hormones. Thyroid follicles of Nile tilapia are observed in stage 21 to stage 22 (Morrison *et al.* 2001), while in other tilapia (*O. mossambicus*), thyroid hormones begin to increase just before the yolk absorption (Reddy *et al.* 1992).

Use of tilapia developmental staging system as a standard for evolutionary developmental biology in haplochromine cichlids

Cichlids are geographically distributed over an essentially Gondwanan pattern (Stiassny 1991), namely, in Asia (India, Madagascar, Sri Lanka), the Americas

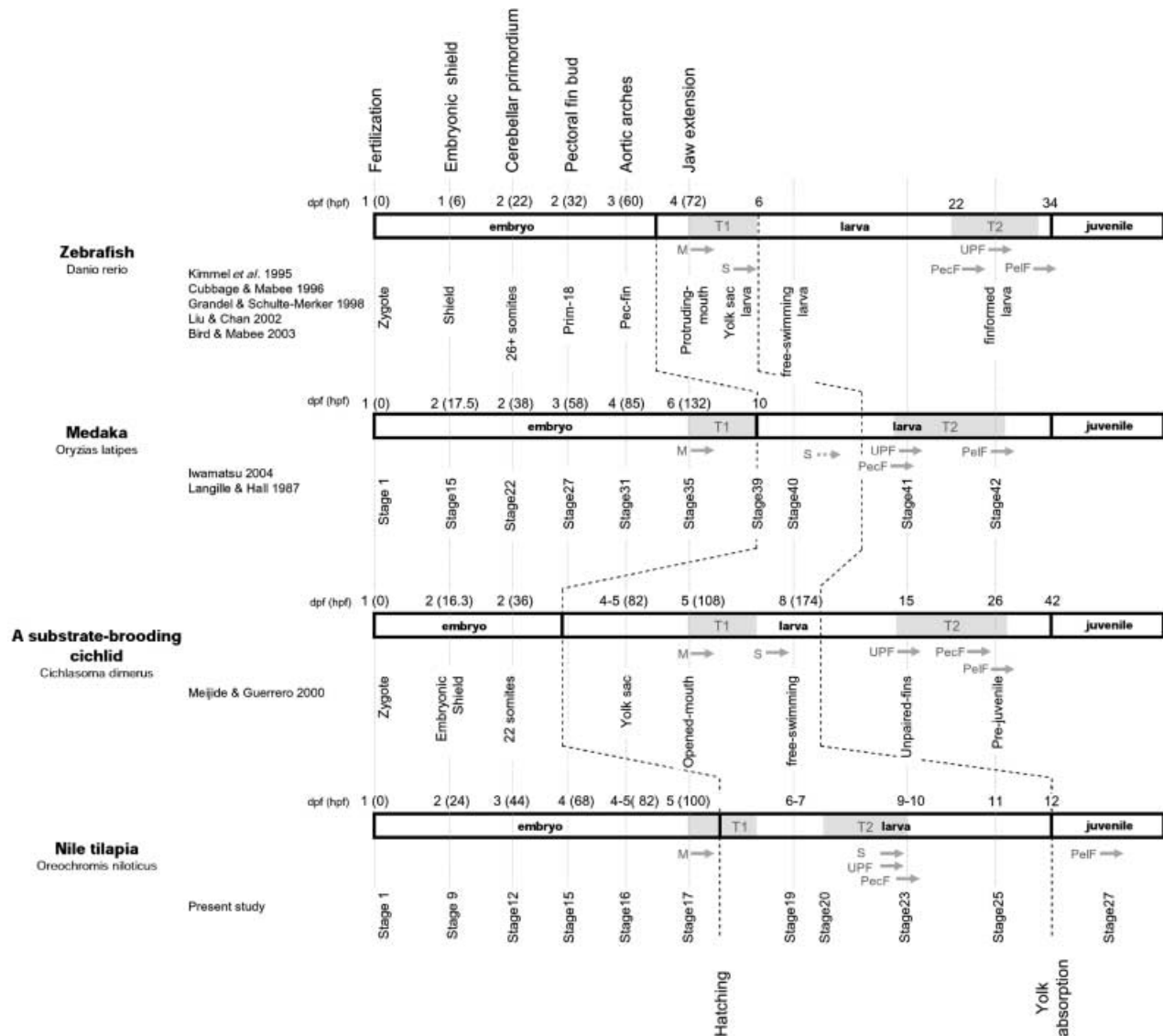


Fig. 10. Tilapia in evolutionary developmental biology of fish. The development of Nile tilapia *O. niloticus* (a mouth-brooder cichlid) is compared with that of other fresh-water fish (two model fish, zebrafish *Danio rerio* and medaka *Oryzias latipes*, and a substrate-brooder cichlid *Cichlasoma dimerus*). (Zebrafish: Kimmel *et al.* 1995; Cabbage & Mabee 1996; Grandel & Schulte-Merker 1998; Liu & Chan 2002; Bird & Mabee 2003. Medaka: Iwamatsu 2004; Langille & Hall 1987. *Cichlasoma dimerus*: Meijide & Guerrero 2000. Nile tilapia: present study.) T1 and T2: 'embryonic to larval' and 'larval to juvenile' transitions of zebrafish (Liu & Chan 2002) and the transitions of the other fish are presumed in view of the morphological changes. dpf, day(s) postfertilization; hpf, hour(s) postfertilization; M, protrusion of mouth; PecF, differentiation of pectoral fins; PelF, differentiation of pelvic fins; S, inflation of swimbladder; UPF, differentiation of unpaired fins.

(Texas in the USA, Central and South America) and Africa (Nelson 2006). Figure 11A shows the phylogenetic relationships among cichlids of the world (Farias *et al.* 2000 and Sparks 2004). Figure 11B shows a general outline of the evolution of African cichlids (Mayer *et al.* 1998; Kocher 2004; Joyce *et al.* 2005). Genus *Oreochromis* is included in the 'Pan-Africa' lineage and is considered to be ancestral to East African cichlids (Klett & Meyer 2002) (Fig. 11B).

Genus *Haplochromis* and the closely related genera, which comprise the species flocks of Lakes Malawi and Victoria, are referred to, conveniently, as the haplochromine cichlids (Greenwood 1981) (Fig. 11B). Recent molecular phylogenetic studies analyzed the evolution of these haplochromine cichlids (Nagl *et al.* 2000; Salzburger *et al.* 2005). The haplochromine cichlids underwent several independent radiations after they colonized Lakes Malawi and Victoria over

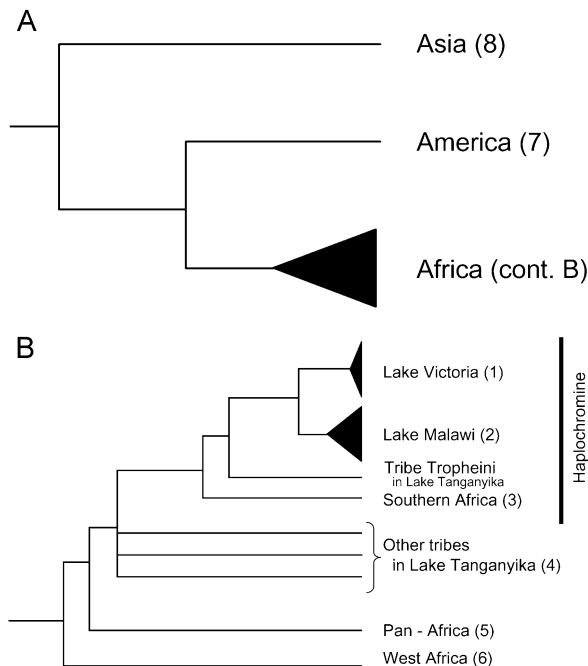


Fig. 11. Phylogenetic relationships of cichlids. (A) Phylogenetic relationship of cichlids in the world as modified from Farias *et al.* (2000) and Sparks (2004); 'cont. B' indicates that the African lineage is expanded in panel B. (B) Phylogenetic relationship among African cichlids as modified from Mayer *et al.* (1998), Kocher (2004) and Joyce *et al.* (2005). The numbers 1–8 in this figure correspond with 'Distribution' in Table 2. Genus *Oreochromis* is included in the 'Pan-Africa' lineage.

the past several million years, and evolutionary biologists have been fascinated by the explosive adaptive evolution exhibited by the haplochromine cichlids (Kocher 2004).

Previously, there were studies on the ontogeny of various cichlids (Table 2). Cichlids are typically categorized into two groups based on their parental care: substrate-brooders, which display custodial care and guarding of fertilized eggs and developing young; and mouth-brooders, which carry the fertilized eggs and developing young within the buccal cavity (Barlow 1991; Keenleyside 1991). Most African cichlids are mouth-brooders, whereas cichlids in the tribe Lamprologini and those in the genus *Boulengerochromis* in Lake Tanganyika are substrate-brooders. Most neotropical (American) cichlids are also substrate-brooders. In general, mouth-brooders have been described as undergoing precocial development, whereas substrate-brooders have been described as altricial (Noakes 1991). All of the haplochromine cichlids are maternal mouth-brooders and Nile tilapia *O. niloticus* is also included in this category.

From the ontogenic point of view, it is interesting to note that the Mozambique tilapia *O. mossambicus*

and the southern mouth-brooder *Pseudocrenilabrus philander* ('Southern Africa' lineage in Fig. 11B) have a similar rate and pattern of development (Holden & Bruton 1994). Also, it was reported that the development of the Lake Malawi cichlids (*Labeotropheus* species, Balon 1977) was similar to that of *O. niloticus*, except that the Lake Malawi cichlids produce fewer, larger eggs (Morrison *et al.* 2001). From the systematic point of view, it has been reported that a group including genus *Oreochromis* was closely affiliated with the tribe Haplochromini (Takahashi 2003). Therefore, we believe that it is appropriate to compare the ontogeny of *O. niloticus* with that of the haplochromine cichlids, despite the phylogenetic differences between tilapia and haplochromine cichlids (~10 million years) (Fig. 11B).

At present, haplochromine cichlids are an emerging model for studies in evolution and development, especially for understanding the genetic and developmental mechanisms that underlie adaptive variation in cichlid trophic morphology (Albertson & Kocher 2001; Albertson *et al.* 2003a,b; Albertson *et al.* 2005; reviewed by Albertson & Kocher 2006). There are also genomic resources and cDNA resources for tilapia and haplochromine cichlids (Kocher *et al.* 1998; Watanabe *et al.* 2003; Renn *et al.* 2004; Katagiri *et al.* 2005; Kijimoto *et al.* 2005; Lee *et al.* 2005; Lang *et al.* 2006). Furthermore, tilapia is a very useful laboratory animal (e.g. a transgenic technique has been developed, reviewed by Maclean *et al.* 2002). There have been classic and detailed papers regarding anatomy and functional morphology using *Haplochromis elegans*, a generalized insectivorous *Haplochromis* (Barel *et al.* 1976; Barel *et al.* 1977; Anker 1978, 1984, 1986, 1987, 1989; Barel 1983). There have also been studies reporting particular events of craniofacial development in cichlids (*Haplochromis elegans*: Verraes & Ismail 1980; Ismail *et al.* 1982; Otten 1982; Huysseune *et al.* 1988; *Hemichromis bimaculatus* and *Astatotilapia burtoni*: Huysseune 1990; *Hemichromis bimaculatus*: Huysseune & Sire 1992a,b).

The evolutionary developmental biology in Darwin's finches (Abzhanov *et al.* 2004) was standardized by reference to the chicken staging system (Hamburger & Hamilton 1951). Here we propose that our developmental staging system of Nile tilapia *O. niloticus* can be used to understand not only the ontogeny of tilapia, but also the evolutionary developmental biology of haplochromine cichlids. For the future of comparative developmental analysis in the haplochromine cichlids, it is necessary for cichlid researchers to utilize standardized laboratory conditions to raise fish (e.g. growth protocol, light cycle, temperature) and share molecular developmental biology data for a particular species. We especially note that a uniform temperature

Table 2. Previous studies of cichlid development

Distribution	Mouth-brooder (M)		
Species name	Substrate-brooder (S)	Author	Year
1. Lake Victoria basin			
<i>Haplochromis elegans</i>	M	Otten	1982
2. Lake Malawi			
<i>Labeotropheus</i> sp.	M	Balon	1977
3. Southern Africa			
<i>Pseudocrenilabrus philander</i>	M	Holden and Bruton	1994
4. Lake Tanganyika			
<i>Tanganicodus irsacae</i>	M	Kuwamura <i>et al.</i>	1989 [†]
<i>Haplotaxodon microlepis</i>	M/S	Kuwamura	1988 [†]
<i>Boulengerochromis microlepis</i>	S	Kuwamura and Mihigo	1988
5. Pan-Africa			
<i>Oreochromis niloticus</i>	M	Nussbaum and Chervinsky	1968
		Galman	1980 [†]
		Lingling and Qianru	1981
		Galman and Avtalion	1989
		Rana	1990
		Morrison <i>et al.</i>	2001
<i>Oreochromis mossambicus</i>	M	Holden and Bruton	1992
<i>Tilapia leucosticta</i>	M	Welcomme	1967 [‡]
<i>Tilapia macrocephala</i>	M	Shaw and Aronson	1954
6. West Africa			
<i>Hemichromis bimaculatus</i>	S	Jones	1972
		Hoepfel	1913 [‡]
		Konnertz	1913 [‡]
		McEwen	1930 [‡]
7. Americas			
<i>Aequidens latifrons</i>	?	Balon and Flank	1953 [‡]
<i>Aequidens pulcher</i>	S	Jones	1972
<i>Cichlasoma bimaculatum</i>	S	Jones	1972
<i>Cichlasoma biocellatum</i>	S	Jones	1972
<i>Cichlasoma dimerus</i>	S	Meijide and Guerrero	2000
<i>Cichlasoma dovii</i>	?	Cabrera <i>et al.</i>	1988 [§]
<i>Cichlasoma istlanum</i>	?	Contreras and Dias	1990 [§]
<i>Cichlasoma nigrofasciatum</i>	S	Balon	1960 [‡]
		Jones	1972
		Martinez and Murillo	1987 [§]
<i>Herichthys cyanoguttatus</i>	?	Balon	1959 [‡]
8. Asia			
<i>Etroplus maculatus</i>	?	Jones	1937 [‡]
		Alikunhi	1951 [‡]

[†]References cited in Morrison *et al.* (2001); [‡]references cited in Jones (1972); [§]references cited in Meijide and Guerrero (2000); [†]references cited in Kuwamura and Mihigo (1988).

condition is very important for standardization. These approaches will facilitate studies into the molecular basis for the key innovation of the pharyngeal jaws (Liem 1973; Galis 2001) and the parallel evolution among cichlids of East African lakes (Kocher *et al.* 1993).

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